Combining Multiangular, Polarimetric, and Hyperspectral Measurements to Estimate Leaf Nitrogen Concentration From Different Plant Species

Ming Liu, Zhongqiu Sun, Shan Lu, and Kenji Omasa

***Abstract*— Optical remote sensing is one of the most popular methods for estimating leaf nitrogen concentration (LNC). This nondestructive approach based on reflected intensity measurements has been applied to estimate the variation and distribution of nitrogen concentration in leaf and canopy levels in numerous studies. However, both intensity and polarization are necessary to describe the optical properties of light reflected from leaves and to estimate LNC estimation. In this study, based on the Stokes parameters, the total reflectance, the polarized reflectance, and the nonpolarized reflectance factors (NpRFs) were simultaneously obtained through polarimetric hyperspectral measurements under varied source-viewing geometries in both laboratory and field conditions. Several published hyperspectral indices based on the NpRF showed much better LNC estimation accuracy than those using the total reflectance factor. A clear improvement was found in the viewing directions dominated by specular reflection. Thus, using multi-angular polarimetric hyperspectral measurements not only improves the accuracy of hyperspectral indices on LNC estimation using the NpRF, but also enables the hyperspectral indices to be effective for a wide range of viewing angles. Moreover, polarimetric measurements deepen the understanding of the optical properties of light reflected from leaves. These results indicate that the combination of multiangular, polarimetric, and hyperspectral measurements may play a key role in the estimation of LNC.**

***Index Terms*— Hyperspectral indices, leaf nitrogen concentration (LNC), multiangular measurements, nonpolarized reflectance factor (NpRF), polarimetric properties.**

I. Introduction

NITROGEN is the chemical element in the leaf that mostly limits plant growth and productivity, affecting the functioning and dynamics of forested and agricultural ecosystems [1]–[5]. Understanding the variation and distribution of leaf nitrogen concentration (LNC) at different scales thus provides insights for various ecological and agricultural studies [6]–[9]. Optical remote sensing has been widely used to estimate LNC at leaf, plant, and stand levels that can be related to landscapes, regions, and continents [10]–[18]. Compared with traditional measurement methods, optical remote-sensing approaches allow nondestructive and repeatable LNC estimation, which can provide continuous temporal and spatial distributions of nitrogen concentration that can be used to improve predictions of crop yields and net primary production [16], [19]–[23].

Plant leaf nitrogen estimations using optical measurements are based on change in leaf spectral reflectance in response to growth conditions controlled by nitrogen concentration [14], [24]. For example, leaf nitrogen is associated with the presence of chlorophyll, which varies the reflectance in the visible part of the spectrum and is particularly sensitive to radiation in the red edge [25]–[28]. In practice, most studies use the intensity (defined as the hyperspectral reflectance or reflectance factor) derived from photometric measurements to estimate LNC based on empirical models (including hyperspectral indices and machine-learning methods [22], [29], [30], [100]), physically based models [11], [35], [36], [101], or hybrid regression methods [100], [102]. Such models have been applied to ground (laboratory and field), airborne, and spaceborne measurements [30]–[32], [34], [100]. In general, the success of a remote-sensing approach depends on the understanding of factors affecting the optical properties of leaves and how these extend to canopy level [30], [31], [35], [36].

In fact, both intensity and polarization are necessary to describe the optical properties of light reflected from leaves [37]. Polarimetric measurements have been first used to quantify the optical properties of leaves [38]–[40] and then used to characterize the structure of the plant canopies [41]–[45]. The clear relationship between polarization properties and the architecture of the vegetation canopy is due to the polarized signals that are primarily generated from the leaf surface [46]–[49]; the leaf surface properties and the distribution of leaf orientation control the polarization

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TABLE I

STATISTICAL RESULTS OF LNC IN SIX TYPES OF PLANTS ARE MEASURED IN THE LABORATORY AND IN THE FIELD; 176 CALIBRATION SAMPLES WERE RELATED TO HYPERSPECTRAL INDICES AND 86 LEAVES MADE UP THE VALIDATION SAMPLES. CV REPRESENTS THE COEFFICIENT OF VARIATION

	Species	Number of samples		LNC_{max}(%)	LNC_{min}(%)	LNC_{means}(%)	CV
		Laboratory	Field				
Calibration	Sche	25	5	3.74	0.45	2.27	0.41
	Pach	19	22	4.20	1.24	2.98	0.28
	Jugl	11	24	4.02	1.00	2.68	0.33
	Epip	23	0	4.13	0.86	2.46	0.42
	Acer	22	4	4.11	0.91	2.49	0.45
	Popu	16	5	3.33	0.96	2.26	0.35
	Total	116	60	4.20	0.45	2.57	0.37
Validation	Sche	10	0	3.40	1.60	2.42	0.24
	Pach	11	7	4.23	1.25	3.03	0.31
	Jugl	6	11	4.34	1.33	3.02	0.30
	Epip	12	0	3.90	1.00	2.40	0.45
	Acer	14	3	4.04	1.09	2.51	0.42
	Popu	6	6	3.31	0.92	2.17	0.41
	Total	59	27	4.34	0.92	2.65	0.37

properties for different plant canopies [41], [50]–[52]. Based on the Fresnel formula, researchers have proposed several polarized reflectance models for quantifying the polarization properties of plants and further clarified that the polarized portion can be directly related to the specular reflection from the leaf surface [38], [42], [47], [49], [53]–[56]. These findings deepened our understanding of the properties of light reflected from vegetated surfaces.

In the estimation of leaf biochemical parameters, such as LNC, the fraction of specular reflection from leaf surfaces measured by the sensors may weaken the sensitivity of the reflected signals to leaf interior architecture and composition [57], [58]. This is because the specular reflection from the leaf surface only depends on the leaf surface structural properties [59]–[61] and is independent of the constitution of the leaf tissue [38], [62]. A physical model, PROCOSINE [11], has been used to account for the specular component of light reflected from leaves based on the PROSPECT [36]. This model further improves our understanding of the effects of specular reflection on estimating leaf biochemical parameters [11]. Because the polarized reflectance factor is mainly derived from the specular reflection from the leaf surface, researchers concluded that the nonpolarized proportion is indistinguishable from the diffuse reflectance, which has the potential to relate to leaf inner composition [62]. However, researchers still have not investigated the possibility of directly using nonpolarized spectral signals to estimate LNC, even though the polarization properties of plant leaves, combined with hyperspectral and other imaging measurements, have been explored by several studies [44], [49], [51], [63]–[66].

In this study, using multiangular, polarimetric, and hyperspectral measurements of leaves with different surface properties from six plant species in both laboratory and field conditions, the Stokes parameters were used to quantify the total reflectance factor, polarized reflectance factor, and nonpolarized reflectance factor (NpRF) of leaves. The primary goal was to study the hyperspectral indices (one of the most commonly used methods) for LNC estimation to understand whether we can rely on non-polarized hyperspectral reflectance factors to improve LNC estimation accuracy. Moreover, we also investigated whether the total reflectance factors derived from photometric and polarimetric measurements yield to similar LNC estimates using hyperspectral indices. These parameters obtained from polarimetric measurements may greatly simplify the measurement efforts. This study aims to demonstrate the potential of using multi-angular, polarimetric, and hyperspectral measurements for estimating LNC across different plant species.

II. SAMPLES AND METHODS

A. Leaf Samples

In this study, a total of 262 samples from six different plant species [*Schefflera microphylla Merr.* (Sche, $n = 40$), *Pachira macrocarpa* (Pahe, $n = 59$), *Juglans* (Jugl, $n = 52$), *Epipremnum aureum* (Epip, $n = 35$), *Acer saccharum Marsh* (Acer, $n = 43$), and *Populus L.* (Popu, $n = 33$)] were collected for multiangular photometric and polarimetric measurements, of which 175 samples were measured in the laboratory and 87 samples were measured in the field. We used 116 laboratory samples, 60 field samples in the calibration dataset, and 86 samples in the validation dataset (Table I). It should be noted in this study that we did not analyze the leaf surface structure, such as the refractive index or roughness, which can influence the distribution of reflectance factors of leaf [59], [60].

B. Definition of Polarimetric and Photometric Properties of Leaves

The reflected light from the leaf surface is partly polarized when the incident light is nonpolarized or natural [37]. Then the polarimetric properties of light reflected from leaf surface can be characterized by the Stokes parameters, which derived

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from four linearly polarized radiance of leaf [37], [38], [48]. The Stokes parameters (I, Q, U, and V) are as follows:

$$I = (L_{0^\circ} + L_{45^\circ} + L_{90^\circ} + L_{135^\circ})/2 \quad (1)$$

$$Q = L_{0^\circ} - L_{90^\circ} \quad (2)$$

$$U = L_{45^\circ} - L_{135^\circ} \quad (3)$$

$$R_p = \sqrt{Q^2 + U^2} \quad (4)$$

where I represents the total intensity of radiation reflected from leaves, Q and U represent the magnitude and orientation of the polarization ellipse semimajor axis, respectively, and the fourth Stokes parameter, V, is negligible after reflection from the leaves [37]. L_{0°, L_{45°, L_{90°, and L_{135° are the polarized radiance in different polarizer directions (0°, 45°, 90°, and 135°) from the leaf. The meridional plane of the instrument is used as the reference plane for the polarizer during polarimetric measurements, and R_p is the polarized radiance of the leaf sample. To represent the intensity reflected from the leaf surface based on the Stokes parameters, the I parameter reflectance factor (IpRF) was used. The IpRF is defined as the ratio of the sum of the polarized radiance [(1)] in four polarized states (dI_{sample}) from the leaf surface to the reflected radiance ($dL_{\text{reference}}$) from a diffuse reference surface (in this case, the Spectralon) under the same source-viewing geometries

$$\text{IpRF}(\lambda, \theta_s, \theta_v, \phi_s, \phi_v) = \frac{dI_{\text{sample}}(\lambda, \theta_s, \theta_v; \phi_s, \phi_v)}{dL_{\text{reference}}(\lambda, \theta_s, \theta_v; \phi_s, \phi_v)}\rho_\lambda \quad (5)$$

where θ_s is the source zenith angle (SZA) of artificial light, θ_v is the viewing zenith angle (VZA), φ_s is the source azimuth angle, φ_v is the viewing azimuth angle, λ is the wavelength, and ρ_λ is a hemispherical spectral reflectance of the Spectralon panel, which is considered to be the reference panel.

The bidirectional polarized reflectance factor (BPRF) can be used to define the polarization properties of light reflected from the leaf surface; it is calculated as the ratio of the polarized radiance (dR_p) [(4)] from the leaf surface to the reflected radiance ($dL_{\text{reference}}$) from an ideal diffuse reference surface under the same source-viewing geometries

$$\text{BPRF}(\lambda, \theta_s, \theta_v, \phi_s, \phi_v) = \frac{dR_p(\lambda, \theta_s, \theta_v; \phi_s, \phi_v)}{dL_{\text{reference}}(\lambda, \theta_s, \theta_v; \phi_s, \phi_v)}\rho_\lambda. \quad (6)$$

The bidirectional reflectance factor (BRF_{lab}) is used to characterize the photometric measurements of leaves in the laboratory. BRF_{lab} is defined as the ratio of the reflected radiance ($dL_{\text{lab,sample}}$) from the leaf surface to the reflected radiance ($dL_{\text{lab,reference}}$) from an ideal diffuse reference surface under the same source-viewing geometries [67]

$$\text{BRF}_{\text{lab}}(\lambda, \theta_s, \theta_v, \phi_s, \phi_v) = \frac{dL_{\text{lab,sample}}(\lambda, \theta_s, \theta_v; \phi_s, \phi_v)}{dL_{\text{lab,reference}}(\lambda, \theta_s, \theta_v; \phi_s, \phi_v)}\rho_\lambda. \quad (7)$$

Based on the statement in the previous study, if field measurements are taken under clear sky conditions with predominantly direct radiation and small field of view of the sensor, the measurements closely approximate BRFs [99]. Because our field measurements were performed in clear sky condition, $\text{BRF}_{\text{field}}$ is approximately used to describe the photometric measurements of leaves

$$\text{BRF}_{\text{field}}(\lambda, \theta_s, \theta_v, \phi_s, \phi_v) = \frac{dL_{\text{field,sample}}(\lambda, \theta_s, \theta_v; \phi_s, \phi_v)}{dL_{\text{field,reference}}(\lambda, \theta_s, \theta_v; \phi_s, \phi_v)}\rho_\lambda. \quad (8)$$

In theory, the total reflectance factors derived from polarimetric (IpRF) and photometric (BRF_{lab} or $\text{BRF}_{\text{field}}$) measurements should be similar in the laboratory and field. However, when performing the polarimetric measurements, a polarizer is fixed in front of the fiber-optic cable of the spectrometer. Because of the absorption of the polarizer (it does not have a perfect transmission), the polarized radiance in different polarization states is reduced. In order to extract the total reflectance information from the polarimetric measurement, it is necessary to compensate the measured polarized radiance in the four polarizer directions. Thus, the I parameter shown in (1) should be redefined as

$$I_{\text{true}} = \left(\frac{L_{0^\circ}}{1 - \text{abs}_p} + \frac{L_{45^\circ}}{1 - \text{abs}_p} + \frac{L_{90^\circ}}{1 - \text{abs}_p} + \frac{L_{135^\circ}}{1 - \text{abs}_p}\right) \Big/ 2 \quad (9)$$

where abs_p is the polarizer absorption ratio determined by the Spectralon panel

$$\text{abs}_p = 1 - L_{\text{spectralon,polarizer}}/L_{\text{spectralon}} \quad (10)$$

where $L_{\text{spectralon}}$ is the radiance of the Spectralon without a polarizer. The average polarized radiance of the Spectralon in four polarizer directions (0°, 45°, 90°, or 135°) was used to calculate $L_{\text{spectralon,polarizer}}$, $L_{\text{spectralon,polarizer}} = (L_{\text{spectralon},0^\circ} + L_{\text{spectralon},45^\circ} + L_{\text{spectralon},90^\circ} + L_{\text{spectralon},135^\circ})/4$. The detailed calculation of abs_p can be found in [68].

When the true I parameter is calculated using (9), the IpRF in (5) under laboratory and field conditions is redefined as

$$\text{IpRF}_{\text{lab}}(\lambda, \theta_s, \theta_v, \varphi_s, \varphi_v) = \frac{dI_{\text{lab,sample,true}}(\lambda, \theta_s, \theta_v; \varphi_s, \varphi_v)}{dL_{\text{lab,reference}}(\lambda, \theta_s, \theta_v; \varphi_s, \varphi_v)}\rho_\lambda \quad (11)$$

$$\text{IpRF}_{\text{field}}(\lambda, \theta_s, \theta_v, \varphi_s, \varphi_v) = \frac{dI_{\text{field,sample,true}}(\lambda, \theta_s, \theta_v; \varphi_s, \varphi_v)}{dL_{\text{field,reference}}(\lambda, \theta_s, \theta_v; \varphi_s, \varphi_v)} \times \rho_\lambda. \quad (12)$$

Under the consideration of the absorption ratio of the polarizer, the IpRF_{lab} is similar to BRF_{lab}, and $\text{IpRF}_{\text{field}}$ is similar to $\text{BRF}_{\text{field}}$, then the NpRF ($\text{BRF}_{\text{lab},n-p}$ and $\text{BRF}_{\text{field},n-p}$) can be obtained directly from the difference between IpRF_{lab} ($\text{IpRF}_{\text{field}}$) and BPRF_{lab} ($\text{BPRF}_{\text{field}}$) in both laboratory and field measurements, as follows:

$$\text{BRF}_{\text{lab},n-p} = \text{BRF}_{\text{lab}} - \text{BPRF}_{\text{lab}} = \text{IpRF}_{\text{lab}} - \text{BPRF}_{\text{lab}} \quad (13)$$

$$\text{BRF}_{\text{field},n-p} = \text{BRF}_{\text{field}} - \text{BPRF}_{\text{field}} = \text{IpRF}_{\text{field}} - \text{BPRF}_{\text{field}}. \quad (14)$$

To abbreviate these reflectance quantities, $\text{BRF}_{\text{lab-field}}$ represents the BRF_{lab} and $\text{BRF}_{\text{field}}$, IpRF represents the IpRF_{lab} and $\text{IpRF}_{\text{field}}$, as well as NpRF represents $\text{BRF}_{\text{lab},n-p}$ and $\text{BRF}_{\text{field},n-p}$. Other notations are listed in Table II.

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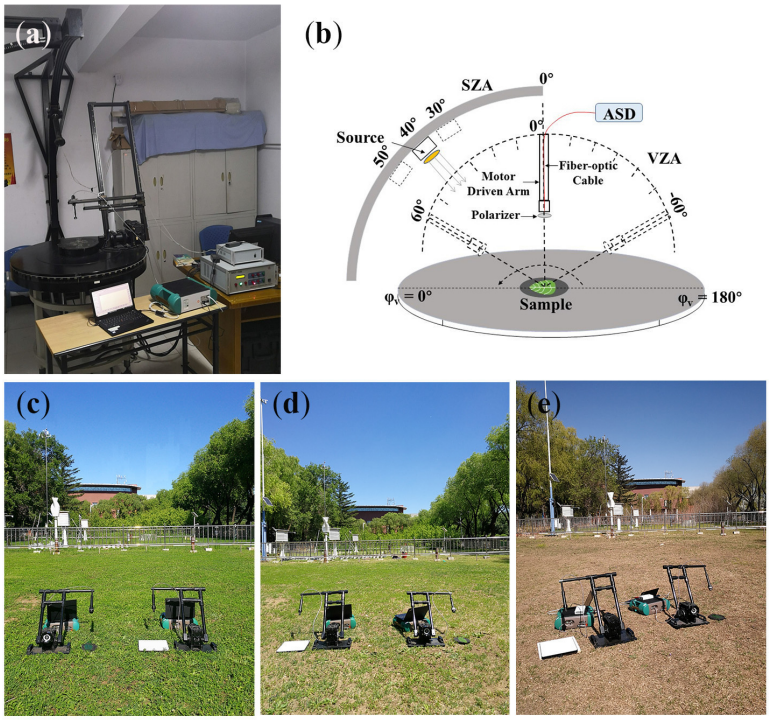


Fig. 1. (a) Laboratory measurements. (b) System diagram of measurements in the principal plane, where $\varphi_v = 180°$ corresponds to the forward scattering direction. SZA and VZA correspond to SZA and VZA, respectively. (c)–(e) Multiangle photometric and polarimetric measurements under different incident conditions in the principal plane in the field. The two ASD spectrometers are used to measure photometric and polarimetric signals from leaves and to measure the reflected radiance from a Spectralon panel with the same measurement geometry, respectively.

TABLE II

NOTATIONS USED FOR THE DESCRIPTION OF ANGULAR PHOTOMETRIC AND POLARIMETRIC MEASUREMENTS

Symbols	Description
L	Reflected radiance (W m^{-2} $sr^{-1}nm^{-1}$)
I	Total reflected radiance from polarimetric measurement (W m^{-2} $sr^{-1}nm^{-1}$)
Q	Magnitude of the polarization ellipse semimajor axis
U	Orientation of the polarization ellipse semimajor axis
θ	Zenith angle (incident or viewing) (°)
φ	Azimuth angle (°)
λ	Wavelength (nm)
ρ	Hemispherical spectral reflectance of the Spectralon panel
abs	The absorption ratio of the polarizer
Subscripts	
0°, 45°, 90° and 135°	Polarized radiance at different polarizer directions
p/polarizer	Polarized radiance
lab	Laboratory
field	Field
sample	Leaf
reference/spectralon	Reflected radiance from the Spectralon
n-p	Non-polarized proportion of total reflection
true	Compensated the measured reflected radiance
terms	
BRF	Bidirectional reflectance factor
BPRF	Bidirectional polarized reflectance factor measured in lab and field
IpRF	*I* parameter reflectance factor measured in lab and field
NpRF	Non-polarized reflectance factor measured in lab and field
SZA	Source zenith angle (°)
VZA	Viewing zenith angle (°)
$BRF_{lab\text{-}field}$	The combination of bidirectional reflectance factor (BRF) in both laboratory and field

C. Instruments and Measurements

In this study, the laboratory multiangular photometric and polarimetric hyperspectral characteristics of the leaves from six plant species were measured using the Northeast Normal University Laboratory Goniospectrometer System (NENULGS) [69] in Fig. 1(a). The NENULGS consists of three main parts: a goniometer, an artificial

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illumination source, and a spectroradiometer [Analytical Spectral Devices (ASD) FieldSpec 4, Colorado, USA]. The goniometer can change the viewing direction with an accuracy of 0.1° in the range 0°–90°. The artificial illumination source is a tungsten halogen lamp, and the incident light is nonpolarized [69]. The calcite Gran Thompson prism is used to measure the polarization characteristics of plant leaves.

In laboratory multiangular polarimetric and photometric measurements, the SZAs were set as 30°, 40°, and 50°. The single leaf was measured in the principal plane and the interval of the VZA was 10° at the VZAs from −60° to 60° (the minus sign "−" corresponds to the forward scattering directions) in Fig. 1(b). Because of the structural limitation of the NENULGS, the VZA that is equal to SZA in the backward scattering direction cannot be measured. The polarimetric and photometric measurements took about 15 min, and we assumed that leaf biochemical properties were not significantly disturbed by the artificial light.

In field measurements, two similar goniometers were combined with two ASD spectroradiometers, which were calibrated before the photopolarimetric measurements. The incident zenith angles ranged from 23.0° to 55.7°. Multiangular photometric and polarimetric measurements were performed in the principal plane. One goniometer and one ASD spectroradiometer measured the photometric and polarimetric reflected radiance from leaves and the other goniometer and ASD spectroradiometer measured the reflected radiance from a reference panel, as shown in Fig. 1(c)–(e). Both goniometers had the same illumination-viewing geometries, and both leaf surface and reference panel had the same distance to the sensors. In order to avoid the changes in leaf biochemical parameters under strong sun light, the maximum number of VZA was five for some leaves; two, three, or four VZAs were measured for other samples. The measurement of one leaf sample took less than 3 min.

For both laboratory and field measurements, the field of view of the sensor was 8°, and the distance from the sensor to the sample was 20 cm. When the VZA changed from 0° to 60°, the measured surface area varied from circular to elliptical. The diameter was 2.8 cm at nadir in one leaf sample per acquisition, and it would change to 5.8 cm (the main axis of the ellipse) at the max off-nadir zenith angle. The leaf sample area was larger than the measured area during the measurements.

D. LNC Measurements

After measuring the polarimetric and photometric characteristics of the leaves, each sample was immediately oven-dried at 105 °C for 30 min and then dried to a constant weight at 80 °C for 36 h. The Dumas combustion nitrogen determination method (EA3000 Automatic Elemental Analyzer, Italy) was used to analyze the nitrogen concentration of different leaves [70]. Based on the strict definition [98], the leaf nitrogen was quantified using LNC (LNC, nitrogen mass per leaf dry mass, N%) in this study. The LNC of each plant was the mean of three measurements. Statistical results of LNC for both the laboratory and field are shown in Table I.

TABLE III

HYPERSPECTRAL INDICES USED IN THIS STUDY. D: DERIVATIVE INDICES; SR: SIMPLE RATIO INDICES; NR: NORMALIZED DIFFERENCE INDICES; LOG: LOGARITHMIC INDEX; AND OF-IS: OTHER FORMS OF INDICES

Type	Number	Hyperspectral index	Source
D	(1)	D_{720}	[71]
	(2)	D_{725}	[72]
SR	(3)	R_{740}/R_{720}	[73]
	(4)	R_{750}/R_{710}	[74]
	(5)	R_{747}/R_{708}	[75]
	(6)	R_{550}/R_{800}	[76]
	(7)	R_{750}/R_{550}	[77]
	(8)	R_{750}/R_{700}	[27]
ND	(9)	$(R_{762}-R_{527})/(R_{762}+R_{527})$	[78]
	(10)	$(R_{750}-R_{550})/(R_{750}+R_{550})$	[79]
	(11)	$(R_{790}-R_{720})/(R_{790}+R_{720})$	[80]
	(12)	$(R_{750}-R_{705})/(R_{750}+R_{705})$	[27]
LOG	(13)	$LOG(1/R_{720})'$	[81]
OF-Is	(14)	$(R_{780}-R_{710})/(R_{780}-R_{680})$	[82]
	(15)	$(R_{750}-R_{705})/(R_{750}+R_{705}-2*R_{445})$	[83]
	(16)	$(R_{755}+R_{680}-2*R_{705})/(R_{755}-R_{680})$	[84]

E. Hyperspectral Indices

Several types of hyperspectral indices were used in this study (Table III): derivative indices (D), simple ratio indices (SR), normalized difference indices (ND), logarithmic index (LOG), and other forms of indices (OF-Is). Some of them were directly used to estimate LNC, and some were used to estimate leaf chlorophyll content (LCC), which can be related to LNC in a certain extent [71], [72]. The relationships between these hyperspectral indices and LNC were investigated based on the total reflectance factors ($IpRF_{lab}$, $IpRF_{field}$, BRF_{lab}, and BRF_{field}) derived from polarimetric and photometric measurements and NpRF ($BRF_{lab,n-p}$ and $BRF_{field,n-p}$). The coefficient of determination (R^2) was used to explain the relationship between hyperspectral indices and LNC. The root mean square error (RMSE) was used to calculate the deviation between measured LNC and estimated LNC.

F. PROCOSINE Model

The Fresnel equation is the physical theory for explaining the intensity of specular component from leaf surfaces based on the sum of squares of perpendicular and parallel amplitude reflection coefficients [38], [47]. The difference between the perpendicular and parallel amplitude reflection coefficients in the Fresnel equation can also get the polarization after the light reflects from the leaf surfaces [48]–[50]. Thus, the specularly reflected light is partially polarized, and polarization can account for the specular component of leaf surfaces [37]–[39]. PROCOSINE has been used to estimate the leaf biochemical parameters by considering the specular component of light reflected from leaves [11]. We will use the PROCOSINE to approximately account for the polarization effect on the estimation of leaf chemistry. PROCOSINE is derived from the combination of the PROSPECT and COSINE models [11], [36]. Under the condition that the illumination angle equals to the incident angle [11], PROCOSINE can be

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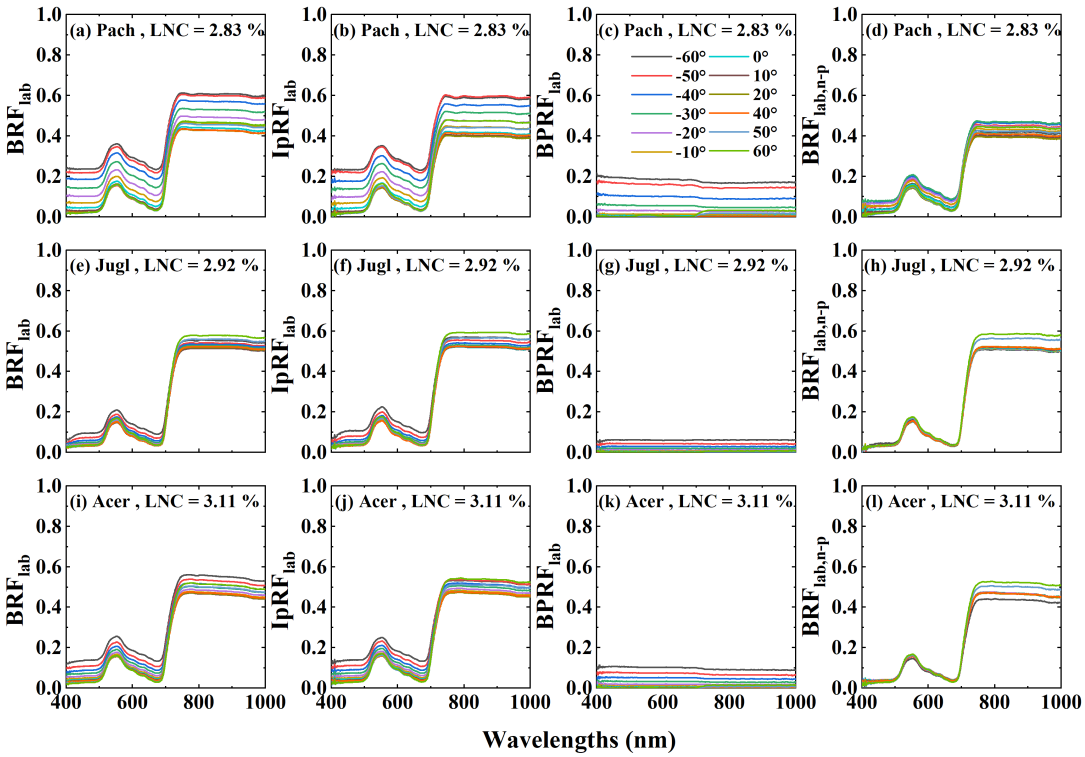


Fig. 2. Laboratory-measured BRF_{lab}, $IpRF_{lab}$, $BPRF_{lab}$, and $BRF_{lab,n-p}$ for plant leaves of different species. First column: BRF_{lab} of leaves at 12 different VZAs in the principal plane. Second column: $IpRF_{lab}$ of leaves. Third column: $BPRF_{lab}$ of leaves at different VZAs. Fourth column: $BRF_{lab,n-p}$ of leaves. The SZV was 40°.

TABLE IV

STATISTICAL CHARACTERISTICS OF EACH PROCOSINE MODEL PARAMETER: LEAF STRUCTURE INDEX (N), LCC, LEAF CAROTENOID CONTENT (CAR), EQUIVALENT WATER THICKNESS (EWT), LEAF MASS AREA (LMA), AND AMOUNT OF SPECULAR REFLECTION (b)

	N	LCC	Car	EWT	LMA	b
Unit		μg/cm^2	μg/cm^2	g/cm^2	g/cm^2	
Minimum	0.90	0.33	0.18	0.0024	0.0018	0
Maximum	3.06	99.81	19.12	0.0625	0.0406	0.60
Mean value	1.59	33.60	8.73	0.0132	0.0079	0.24
Standard deviation	0.30	17.39	3.66	0.0074	0.0037	0.20

simplified as follows [11], [103]:

$$\text{PROCOSINE} = \text{PROSPECT} + b \tag{15}$$

where b is the specular reflection [11], [103]. The statistical characteristics of each PROCOSINE parameter are shown in Table IV. Five levels are selected to represent leaves with different amounts of specular reflection, and the number of spectral reflectance factors was 510 for each specular reflection level. All the biochemical parameters of leaves are the same for each specular reflection level.

III. RESULTS

A. Multiangular Photometric and Polarimetric Hyperspectral Properties of Leaves

A detailed description of the optical properties of lights reflected from leaves requires not only the angular distribution of reflection at different wavelengths, but also the polarized and nonpolarized proportion of refection. Fig. 2 shows the hyperspectral BRF_{lab}, $IpRF_{lab}$, $BPRF_{lab}$, and $BRF_{lab,n-p}$ of leaves from three plant species, Pach., Jugl., and Acer, from laboratory measurements. The BRF_{lab} and $IpRF_{lab}$ (the first and the second columns) have similar angular distribution patterns, with large and angle-sensitive reflectance factors found in the forward scattering directions, and weak anisotropic reflection in the backward scattering direction. Leaves of Pach have the strongest anisotropic BRF_{lab} or $IpRF_{lab}$ property, followed by Acer, and Jugl has a weak anisotropic BRF_{lab} or $IpRF_{lab}$ property. Differences in the angular distribution of reflection [Fig. 2(a), (e), and (i), the first column] from leaves are caused by the different magnitude of surface reflection (or specular reflection) in the forward scattering directions generated by various leaf surface structures [59], [61], [85], [86]. Generally, the smoother the leaf surface, the higher the specular effect on the reflectance factor. In addition, the surface reflection (or specular reflection) primarily dominates the polarized reflectance of leaves [47], [49], [51], [87]. Thus, the stronger the anisotropic BRF_{lab} or $IpRF_{lab}$, the greater the $BPRF_{lab}$ [Fig. 2(c), (g), and (k), the third column].

After the $BPRF_{lab}$ was separated from $IpRF_{lab}$, the anisotropy of the $BRF_{lab,n-p}$ of leaves decreased dramatically [Fig. 2(d), (h), and (l)]. Field measurements (Fig. 3) yielded results that were similar to those in Fig. 2, showing that most of the BRF_{field} and $IpRF_{field}$ of leaves (the first and the second columns in Fig. 3) have stronger anisotropic properties than $BRF_{field,n-p}$ (the fourth column in Fig. 3). In addition, the $BPRF_{lab}$ and $BPRF_{field}$ have

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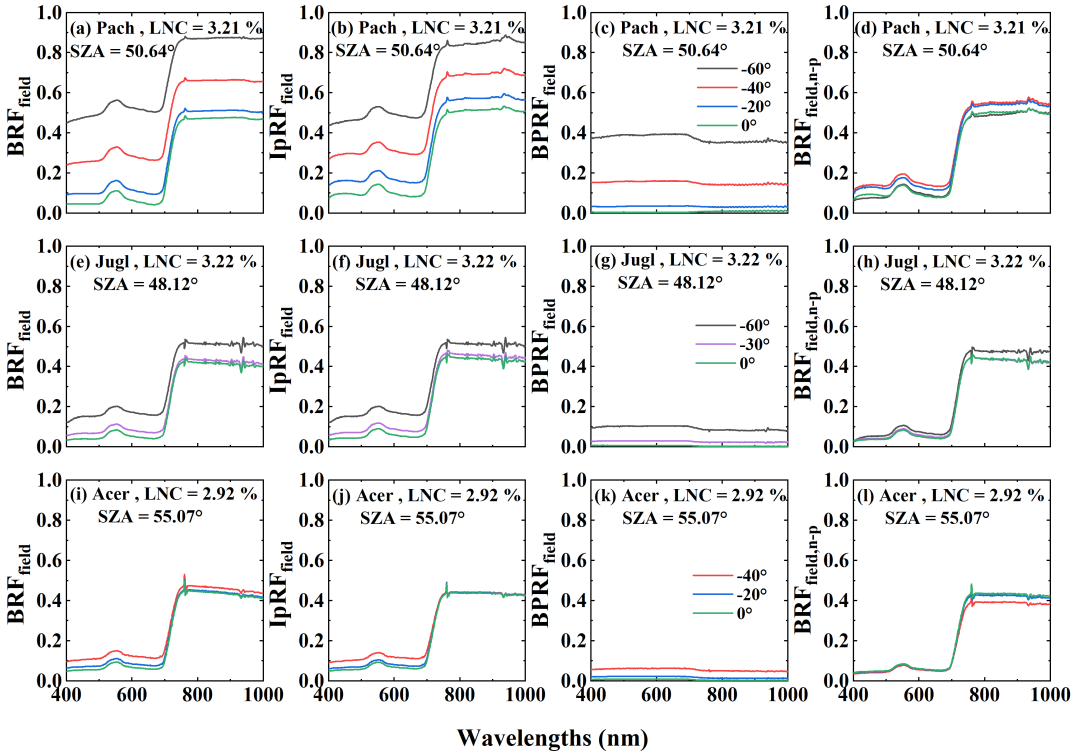


Fig. 3. Field-measured BRF_{field}, $IpRF_{field}$, $BPRF_{field}$, and $BRF_{field,n-p}$ for plant leaves of different species. First column: BRF_{field} of leaves at different VZAs in the principal plane. Second column: $IpRF_{field}$ of leaves. Third column: $BPRF_{field}$ of leaves at different VZAs. Fourth column: $BRF_{field,n-p}$ of leaves. For three plant species, the VZAs were different. (a)–(d) Pach was measured at four different VZAs (-60°, -40°, -20°, and 0°). (e)–(h) Jugl was measured at three different VZAs (-60°, -30°, and 0°). (i)–(l) Acer was measured at three different VZAs (-40°, -0°, and 0°). "-" represents the forward scattering direction.

a weak dependence on wavelengths in both laboratory [Fig. 2(c), (g), and (k)] and field [Fig. 3(c), (g), and (k)]. This is because the leaf surface refractive index, as a function of the wavelength, leads to the variation of BPRF at different wavelengths based on the Fresnel formula [87], [92]. The polarized reflectance factors in this study are consistent with previous results [44], [87].

In Sections III-B and III-C, the combination of laboratory and field hyperspectral reflectance factors $BRF_{lab-field}$ (BRF_{lab} and BRF_{field}), IpRF ($IpRF_{lab}$ and $IpRF_{field}$), and NpRFs ($BRF_{lab,n-p}$ and $BRF_{field,n-p}$) were used to relate to LNC of the calibration leaf samples and then used to estimate LNC of the validation leaf samples.

B. Relationships Between Hyperspectral Indices and LNC Based on $BRF_{lab-field}$, IpRF, and NpRF

The relationships between hyperspectral reflectance factors and LNC from all six plant species (all the calibration samples) in the range from 400 to 1000 nm at different VZAs are shown in Fig. 4. It is clear that NpRF [(13) and (14)] has stronger relationships (higher R^2) to LNC than the $BRF_{lab-field}$ [(7) and (8)] and IpRF [(11) and (12)] in the visible and red-edge wavelengths, and the R^2 is sensitive to VZA when IpRF and $BRF_{lab-field}$ are used in the forward scattering directions [Fig. 4(a)–(e)]. Near the nadir and in the backward scattering directions, the NpRF has the similar R^2 as the $BRF_{lab-field}$ and IpRF [Fig. 4(f)–(m)]. The other interesting finding is that the R^2 from $BRF_{lab-field}$ and IpRF are also similar to each other in the range from 400 to 1000 nm at different VZAs, overlapping at each wavelength. This phenomenon indicates that the intensity measured from photometric measurements (defined as BRF_{lab} or BRF_{field}) can be replaced by the intensity derived from polarimetric measurements (defined as $IpRF_{lab}$ or $IpRF_{field}$) of leaves.

The larger R^2 between NpRF and LNC is found when comparing with IpRF and $BRF_{lab-field}$ for each plant species over all the VZAs (Fig. 5), and these results further confirm that NpRF improves the relationship with LNC. Moreover, the similar R^2 based on $BRF_{lab-field}$ and IpRF (Fig. 5) are also the evidences that IpRF can be used to represent the photometric property of light reflected from leaves for different plant species. Thus, the IpRF, which replaces the reflectance factors (BRF_{lab} and BRF_{field}) in both laboratory and field, was used to represent the photometric property of leaves in the following section.

After confirming that NpRF can improve the R^2 at individual wavelengths in the visible and red-edge range, we calculated the relationships between hyperspectral indices and LNC based on IpRF and NpRF over all the VZAs (Table V). We found that NpRF improves the relationships between most hyperspectral indices and LNC when comparing with IpRF, such as for all the SR, ND, and LOG indices [Fig. 6(a) and (b)].

For the D indices and OF-Is, there was a weak improvement in R^2 when we used the NpRF. This is because R^2_{IpRF} and R^2_{NpRF} for the D and OF-Is are similar at each VZA [Fig. 6(c) and (d)]. Generally, the spectral indices, whose

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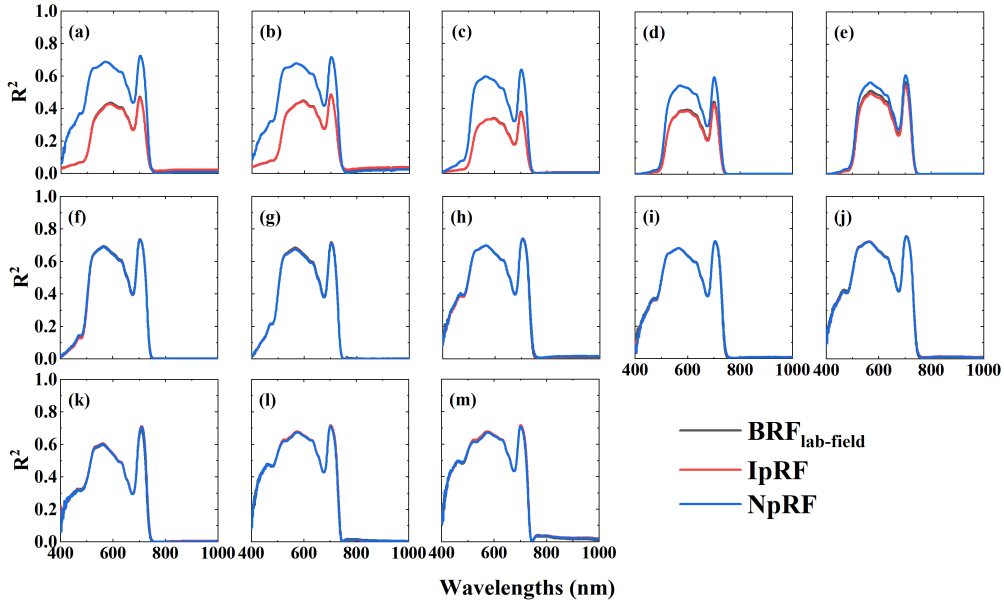


Fig. 4. (a)–(e) Relationships (R^2) between hyperspectral $BRF_{lab-field}$ and LNC at different VZAs in the principal plane, $n = 176$. "−" corresponds to the forward scattering directions. The R^2 values from $BRF_{lab-field}$ and IpRF are very similar, and the NpRF obviously improves the relationships to LNC in the forward scattering directions. The $BRF_{lab-field}$ represents the BRF_{lab} and BRF_{field}, IpRF represents the $IpRF_{lab}$ and $IpRF_{field}$, and NpRF represents $BRF_{lab,n-p}$ and $BRF_{field,n-p}$.

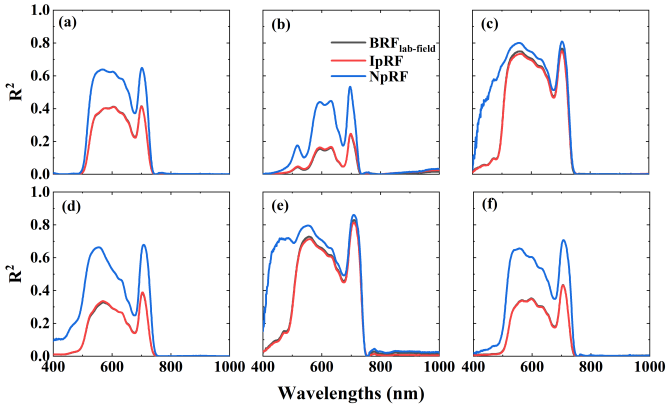


Fig. 5. R^2 between spectral reflectance factors and LNC for each plant species based on the measurements over all the VZAs [(a) Sach, $n = 30$; (b) Pach, $n = 41$; (c) Jugl, $n = 35$; (d) Epip, $n = 23$; (e) Acer, $n = 26$; and (f) Popu, $n = 21$]. The $BRF_{lab-field}$ represents the BRF_{lab} and BRF_{field}, IpRF represents the $IpRF_{lab}$ and $IpRF_{field}$, and NpRF represents $BRF_{lab,n-p}$ and $BRF_{field,n-p}$.

numerator and denominator are in the form of the difference between wavelengths, are considered to be able to reduce the effect of specular reflection.

Thus, for the hyperspectral indices that cannot reduce specular reflection from the leaf surface, such as R_{750}/R_{710}, the NpRF not only improves the relationships between several hyperspectral indices and LNC, but also makes the slopes of the relationships similar at different VZAs in the forward scattering directions when comparing with IpRF [Fig. 7(a) and (b)]; however, for the hyperspectral index that can reduce the specular reflection, such as $(R_{750} - R_{705})/(R_{750} + R_{705} - 2 \times R_{445})$, the similar relationships and slopes are found at each VZA basing on both IpRF and NpRF [Fig. 7(c) and (d)].

TABLE V

R^2 BETWEEN HYPERSPECTRAL INDICES AND LNC BASED ON THE IpRF AND NpRF OVER ALL THE VZAs ($n = 176$). R^2 IN BOLD INDICATES THAT THE NpRF IMPROVES THE RELATIONSHIPS BETWEEN HYPERSPECTRAL INDICES AND LNC [FROM (3) TO (13)], SUCH AS SR, ND, AND LOG INDICES

Hyperspectral indices	Number	R^2_{IpRF}	R^2_{NpRF}
D_{720}	(1)	0.63	0.65
D_{725}	(2)	0.67	0.68
R_{740}/R_{720}	(3)	0.59	**0.64**
R_{750}/R_{710}	(4)	0.54	**0.62**
R_{747}/R_{708}	(5)	0.53	**0.62**
R_{550}/R_{800}	(6)	0.56	**0.66**
R_{750}/R_{550}	(7)	0.41	**0.58**
R_{750}/R_{700}	(8)	0.45	**0.60**
$(R_{762}-R_{527})/(R_{762}+R_{527})$	(9)	0.48	**0.65**
$(R_{750}-R_{550})/(R_{750}+R_{550})$	(10)	0.55	**0.68**
$(R_{790}-R_{720})/(R_{790}+R_{720})$	(11)	0.62	**0.66**
$(R_{750}-R_{705})/(R_{750}+R_{705})$	(12)	0.65	**0.73**
$LOG(1/R_{720})'$	(13)	0.63	**0.72**
$(R_{780}-R_{710})/(R_{780}-R_{680})$	(14)	0.63	0.65
$(R_{750}-R_{705})/(R_{750}+R_{705}-2*R_{445})$	(15)	0.76	0.76
$(R_{755}+R_{680}-2*R_{705})/(R_{755}-R_{680})$	(16)	0.73	0.73

These results indicate that NpRF can improve the relationships between most hyperspectral indices and LNC in the forward scattering directions, which is dominated by the specular reflection from the leaf surface.

C. Improvement of LNC Estimation Using the Nonpolarized Reflectance Factor

In the estimation of LNC, the algorithms of hyperspectral indices were derived from the combination of measurements over all VZAs. As expected, the $RMSE_{NpRF}$ is smaller than $RMSE_{IpRF}$ for all the SR, ND, and LOG indices (bold R^2

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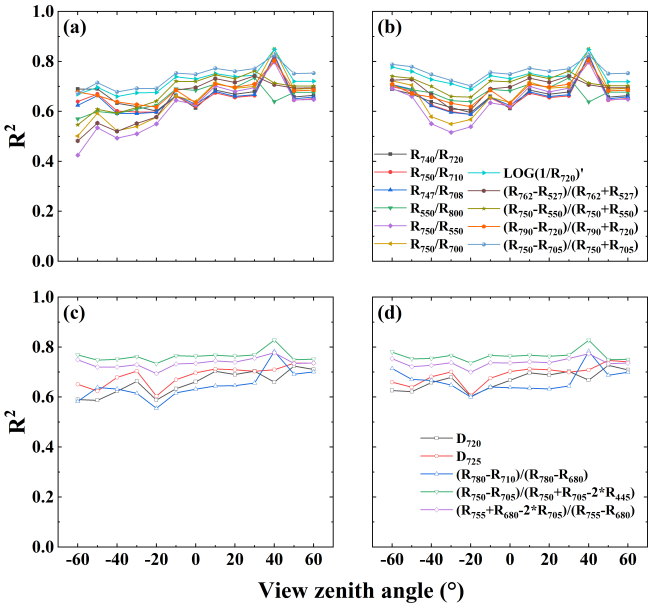


Fig. 6. Comparison between R^2_{IpRF} and R^2_{NpRF} for different indices. These indices, which are in the format of SR, ND, and LOG, have a clear improvement of R^2 based on the comparison between (a) IpRF and (b) NpRF in the forward scattering directions. OF-Is that have a weak improvement of R^2 using (c) and (d) IpRF and NpRF are shown at different VZAs.

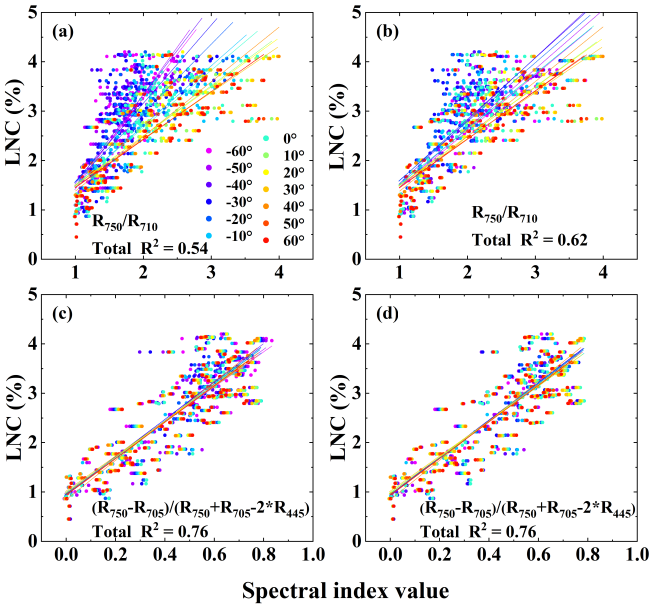


Fig. 7. Relationships between two indices and LNC at different VZAs based on IpRF and NpRF. Lines with different color represent the slope of the relationships between the hyperspectral indices and LNC at different VZAs. The low relationships between R_{750}/R_{710} and LNC and different slopes of the formula are usually found in the forward scattering directions basing on (a) IpRF and improved relationships and similar slopes are found based on (b) NpRF. For $(R_{750} - R_{705})/(R_{750} + R_{705} - 2 \times R_{445})$, similar relationships and slopes are found at each VZA, as shown in (c) and (d).

in Table VI), which indicates that the NpRF improved the ability of hyperspectral indices in estimating LNC when the measurements over all VZAs were used. For the *D* indices and OF-Is, which can reduce the specular reflection from the leaf surface, the improvement is not as clear as for the

TABLE VI

RMSE OF ESTIMATED LNC FOR ALL PLANT SPECIES OVER ALL VZAs, $n = 86$, BASED ON IpRF AND NpRF. SR, ND, AND LOG INDICES [FROM (3) TO (13)] WITH R^2 IN BOLD INDICATE IMPROVED ESTIMATES BY NpRF

Hyperspectral indices	Number	$\text{RMSE}_{\text{IpRF}}$ (N %)	$\text{RMSE}_{\text{NpRF}}$ (N %)
D_{720}	(1)	0.57	0.55
D_{725}	(2)	0.52	0.51
R_{740}/R_{720}	**(3)**	**0.59**	**0.55**
R_{750}/R_{710}	**(4)**	**0.62**	**0.55**
R_{747}/R_{708}	**(5)**	**0.63**	**0.55**
R_{550}/R_{800}	**(6)**	**0.63**	**0.56**
R_{750}/R_{550}	**(7)**	**0.71**	**0.61**
R_{750}/R_{700}	**(8)**	**0.68**	**0.58**
$(R_{762}-R_{527})/(R_{762}+R_{527})$	**(9)**	**0.67**	**0.56**
$(R_{750}-R_{550})/(R_{750}+R_{550})$	**(10)**	**0.64**	**0.55**
$(R_{790}-R_{720})/(R_{790}+R_{720})$	**(11)**	**0.57**	**0.53**
$(R_{750}-R7_{05})/(R_{750}+R_{705})$	**(12)**	**0.57**	**0.50**
LOG$(1/R_{720})$'	**(13)**	**0.58**	**0.51**
$(R_{780}-R_{710})/(R_{780}-R_{680})$	(14)	0.52	0.52
$(R_{750}-R_{705})/(R_{750}+R_{705}-2*R_{445})$	(15)	0.48	0.48
$(R_{755}+R_{680}-2*R_{705})/(R_{755}-R_{680})$	(16)	0.51	0.51

SR, ND, and LOG indices. These results are because NpRF improves the LNC estimation accuracy of all the SR, ND, and LOG indices in the forward scattering directions by reducing the specular effect on the reflection [Fig. 8(b)] compared to using IpRF [Fig. 8(a)]; however, for the D and OF-Is, LNC estimation slightly improved at each VZA when the NpRF was used [see the comparison between Fig. 8(c) and (d)]. All the results shown in Table VI and Fig. 8 confirm that NpRF improves the ability of the hyperspectral indices, which cannot remove the specular reflection, for estimating LNC, especially in the forward scattering directions. Moreover, the NpRF also decreases the dependence of LNC estimation on the VZA for the indices that cannot remove the specular reflection from the leaf surface.

IV. DISCUSSION

A. Testing the Effect of Specular Component on Leaf Biochemical Parameter Estimation Accuracy Using Physical Model

In order to further confirm the influence of the specular reflection on the leaf biochemical parameter estimation accuracy, PROCOSINE simulates the reflectance of leaves with different amounts of specular reflection. Because PROCOSINE cannot account for nitrogen, the indices shown in this study were used to estimate LCC based on these simulated reflectance factors [11]. The simulated spectral reflection of leaves (based on the model parameters in Table VI) with different LCC are shown in Fig. 9, and it is found that the increasing amount of specular reflection increases the reflectance factor in the wavelength range from 400 to 1000 nm.

Basing on the simulated reflectance factor of leaves when $b = 0$, the hyperspectral indices used in this study were related to LCC. Then, the regression model was used to estimate LCC based on the reflectance factor of leaves when *b* was equal to 0 and ranged in 0–0.15, 0.15–0.3, 0.3–0.45, and 0.45–0.6. It is found that the higher specular reflection the larger RMSE

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TABLE VII

RMSE OF ESTIMATED LCC FROM 510 SIMULATED SPECTRAL REFLECTANCE FACTORS FOR EACH SPECULAR REFLECTION LEVEL. THE REGRESSION ALGORITHMS OF HYPERSPECTRAL INDICES WERE DERIVED FROM SIMULATED REFLECTANCE FACTORS WITHOUT CONSIDERING THE SPECULAR REFLECTION ($b = 0$). SR, ND, AND LOG INDICES [FROM (3) TO (13)] IN BOLD INDICATE REDUCED ESTIMATES BY SPECULAR REFLECTION

Spectral Indices	Number	b (specular reflection)				
		0	0-0.15	0.15-0.3	0.3-0.45	0.45-0.6
D_{720}	(1)	8.04	8.04	8.04	8.04	8.04
D_{725}	(2)	6.24	6.24	6.24	6.24	6.24
R_{740}/R_{720}	**(3)**	**3.41**	**8.92**	**17.09**	**21.65**	**24.77**
R_{750}/R_{710}	**(4)**	**3.89**	**10.80**	**19.21**	**23.20**	**25.73**
R_{747}/R_{708}	**(5)**	**3.99**	**11.41**	**19.90**	**23.74**	**26.14**
R_{550}/R_{800}	**(6)**	**7.40**	**12.64**	**23.36**	**30.80**	**36.37**
R_{750}/R_{550}	**(7)**	**3.88**	**15.37**	**23.92**	**26.93**	**28.62**
R_{750}/R_{700}	**(8)**	**4.19**	**14.70**	**22.98**	**25.95**	**27.63**
$(R_{762}-R_{527})/(R_{762}+R_{527})$	**(9)**	**7.26**	**16.81**	**31.00**	**39.01**	**44.36**
$(R_{750}-R_{550})/(R_{750}+R_{550})$	**(10)**	**5.72**	**13.56**	**25.01**	**31.46**	**35.81**
$(R_{790}-R_{720})/(R_{790}+R_{720})$	**(11)**	**3.26**	**8.05**	**15.97**	**20.86**	**24.40**
$(R_{750}-R_{705})/(R_{750}+R_{705})$	**(12)**	**4.50**	**11.07**	**20.93**	**26.61**	**30.52**
$LOG(1/R_{720})'$	**(13)**	**3.74**	**10.23**	**19.49**	**24.70**	**28.26**
$(R_{780}-R_{710})/(R_{780}-R_{680})$	(14)	5.25	5.25	5.25	5.25	5.25
$(R_{750}-R_{705})/(R_{750}+R_{705}-2*R_{445})$	(15)	4.64	4.64	4.64	4.64	4.64
$(R_{755}+R_{680}-2*R_{705})/(R_{755}-R_{680})$	(16)	5.97	5.97	5.97	5.97	5.97

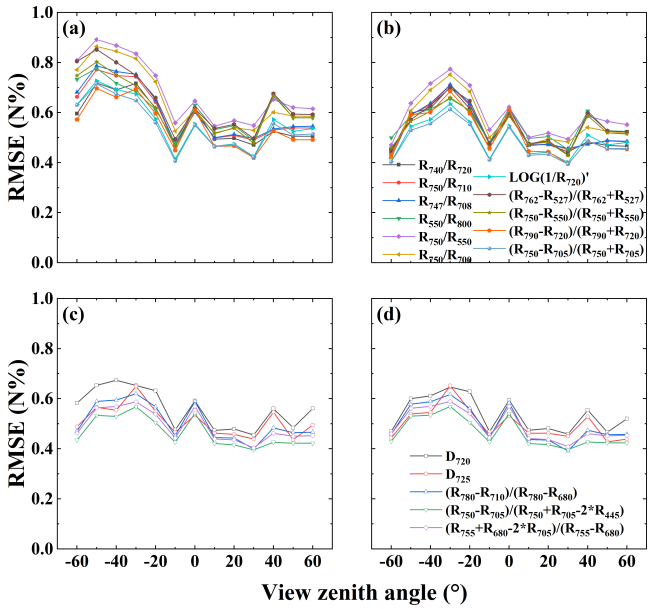


Fig. 8. Comparison of $RMSE_{IpRF}$ and $RMSE_{NpRF}$ for different indices. Indices in the format of SR, ND, and LOG show a clear improvement in LNC estimation based on the comparison of (a) IpRF and (b) NpRF in the forward scattering directions. OF-Is that have a slight improvement of LNC estimation using the IpRF and NpRF are shown in (c) and (d) at different VZAs.

(Table VII) for all the SR, ND, and LOG indices, whereas the LCC estimation accuracy is independent of the amount of specular reflection for the D indices and OF-Is. These results further confirm that the specular reflection reduces the estimation accuracy of leaf biochemical parameters for some indices can reduce the specular reflection effect, which can validate the phenomenon (the higher specular reflection the lower LNC estimation accuracy) corresponding to photometric measurements in the laboratory condition in this study.

Both the measured and modeled results indicate that the simulated specular reflection influences the leaf biochemical parameter estimation accuracy using some hyperspectral indices. Thus, it is necessary to reduce the specular reflection when using the hyperspectral indices (such as SR, ND, and LOG indices) to estimate the leaf biochemical parameters.

B. Significance of Multiangular Polarimetric Measurements of Leaves

Understanding the optical properties of light reflected from leaves is necessary to estimate the biochemical properties' estimates at both leaf and canopy levels [31], [85], [88], [89]. Only when the relationship between hyperspectral indices and LNC was investigated at the leaf level in a detail, it is possible to estimate LNC at the canopy level [30], [31], [90]. This is because several factors influence the ability of hyperspectral reflectance to estimate LNC, such as canopy structure, sun illumination angle, and viewing angle [57], [91].

Polarization as well as intensity have been considered as the essential elements to characterize the optical properties of light reflected from leaves [38], [40], [43], [47], [48], [85], [92]. Generally, the Stokes parameters calculated through the polarized radiance from four or three different polarizer statements, such as I, Q, and U and theirs derivatives [37], were used to define the polarimetric properties of leaves. In the description of the intensity reflected from leaves, reflectance and some of its factors have been used [67]. It seems that if researchers want to measure both polarization and intensity properties of leaves, two different measurements should be performed, one with a polarizer, and the other without a polarizer. In theory, the I parameter calculated from polarimetric measurements equals to the intensity measured by photometric measurements. The consistency between $BRF_{lab-field}$ (BRF_{lab} and BRF_{field}) and IpRF ($IpRF_{lab}$ and $IpRF_{field}$) of leaves when

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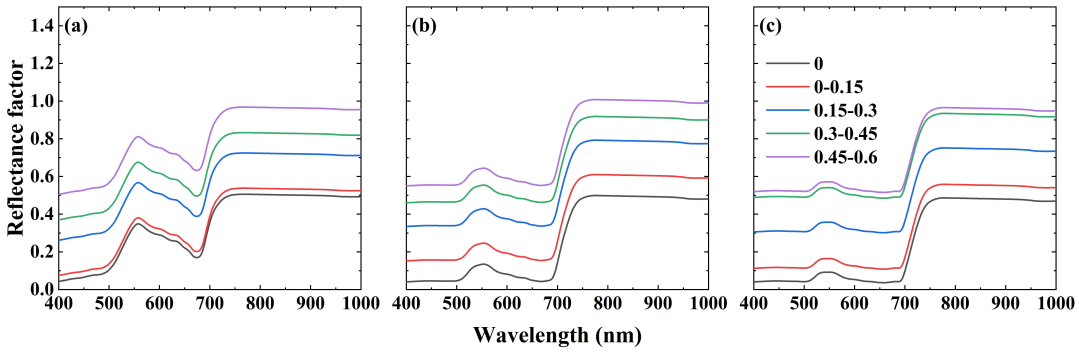


Fig. 9. Simulated spectral reflectance factor of leaves with different amounts of specular reflection (five levels, b = 0, 0–0.15, 0.15–0.3, 0.3–0.45, and 0.45–0.6) using PROCOSINE. (a) LCC = 7.56 μg/cm$_2$. (b) LCC = 37.81 μg/cm$_2$. (c) LCC = 55.35 μg/cm$_2$.

considering the absorption ratio of the polarizer indicate that the intensity derived from polarimetric measurements is the same as that from photometric measurements. Thus, polarimetric measurements can not only describe the photometric property of leaves, but also show the polarization properties of leaves.

In practical applications, measuring polarimetric properties of leaves require more time than photometric measurements, but the IpRF, BPRF, and NpRF of leaves can simultaneously be calculated using polarimetric measurements, and these signals enable researchers to completely understand the optical properties of leaves. On the one hand, IpRF has a similar relationship with LNC as the $BRF_{lab-field}$, and it can be used to estimate LNC in terms of hyperspectral indices. On the other hand, NpRF is directly calculated as the difference between IpRF and BPRF and improves the accuracy of LNC estimation in both laboratory and field conditions. Following the Fresnel equations, angular information is necessary for generating polarized signals of leaves; therefore, the performance of multiangular polarimetric measurements is valuable for understanding the optical properties of leaves for remote-sensing community. Moreover, the improvement of LNC estimation benefits the field of plant science where accurate LNC estimates are needed.

C. Explanation of the NpRF Improves the LNC Estimation at Different Viewing Zenith Angles

BPRF (representing both the $BPRF_{lab}$ and $BPRF_{field}$) is mainly related to leaf surface reflection (or specular reflection), and therefore it does not convey information about leaf biochemical properties. However, few studies assessed the relationships between BPRF and LNC under different measurement conditions. In Fig. 10, it is clear that BPRF at each wavelength in the range 400–1000 nm has very low relationships (R^2 is smaller than 0.035) to LNC at each VZA in the forward scattering directions and over all directions. Thus, the NpRF at each wavelength, which is the difference between IpRF and BPRF, can improve the relationship with LNC at each VZA and over all directions when compared with IpRF. In the backward scattering direction (40° in Fig. 6), there is an R^2 fluctuation for some hyperspectral indices. This is because the number of leaves in this measurement direction

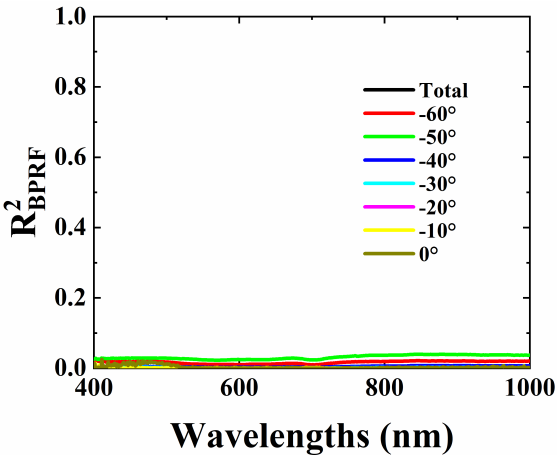


Fig. 10. Relationships between BPRF and LNC at different VZA in the forward scattering directions in the range from 400 to 1000 nm. BPRF corresponds to $BPRF_{lab}$ and $BPRF_{field}$. "Total" represents the combination of all the VZAs shown in this photograph. A very low R^2 was found for each VZA, which indicates that BPRF did not convey the information of LNC in the leaves.

is less than those at other VZAs, and the variability in LNC in this limited sample leads to these less stable R^2 for some hyperspectral indices.

In the forward scattering directions, the specular reflection from the leaf surface (generates the polarized reflectance factor) primarily dominated the total reflectance factor (IpRF), which is responsible for the accuracy of LNC estimation to be clearly dependent on the viewing angles when using the hyperspectral indices in terms of IpRF. In other words, the more the specular reflection from the leaf surface, the more the polarized reflectance factor, and the lower the LNC estimation accuracy. Thus, hyperspectral indices that do not reduce the specular reflection from the leaf surface, such as SR, normalized, and LOG indices, were obviously improved in the forward scattering directions when the NpRF was used. For the D and OF-Is, which can reduce the specular reflection from the leaf surface, the NpRF does not change their LNC estimation accuracy at each VZA. Moreover, the NpRF also decreases the dependence of LNC estimation on VZA [Fig. 8(a) and (b)], which allows to estimate LNC with relative high accuracy in any arbitrary directions.

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Under the consideration of random leaf inclination angle and varied sun incident light, high spatial resolution remote-sensing data, which can measure the reflected signals at centimeter scale, allows to measure the reflection from a part of a single leaf, which may contain much specular information directly reflected from the leaf surface [93]–[97]. Hyperspectral indices, which cannot reduce the specular reflection from the leaf surface, may have low LNC accuracy in this case. The NpRFs derived from the polarimetric measurements enable most types of indices to estimate LNC with stable and considerable accuracy. This advantage from polarimetric measurements makes the hyperspectral indices more effective for LNC estimates when the multiangular measurements are used.

The specular reflection of leaves that exists in the reflected signals of vegetation covers can be obtained by several methods, such as *in-situ* measurements, UAVs (unmanned aerial vehicles), airborne, and spaceborne sensors. However, remote-sensing data at moderate or coarse resolutions (measured using airborne or satellite sensors) includes the leaves in variable directions, whose mean reflectance factors will reduce the contribution of specular reflection to the total reflection and then smooth the specular effects on the LNC estimation. Thus, our proposed polarimetric method may be suitable for the reflected signals obtained via a sensor onboard the UAVs with a high spatial resolution (such as centimeter scale) at a low height when the polarizer can be installed in the front of the sensor.

V. Conclusion

Optical properties of light reflected from leaves with different LNC were described using multiangular, photometric and polarimetric, and hyperspectral measurements in the principal plane under the laboratory and field conditions. The NpRF ($BRF_{lab,n-p}$ and $BRF_{field,n-p}$, which is the difference between the total reflectance factors $IpRF_{lab}$ and $IpRF_{field}$) and the polarized reflectance factors ($BPRF_{lab}$ and $BPRF_{field}$) dramatically improved the ability of the most spectral indices to estimate LNC in the forward scattering directions. This is because polarized reflectance factors, which are derived from the specular reflection of the leaf surface, bear no information about the leaf biochemical parameters. The removal of the polarized reflectance factor, in fact, used the diffused reflection (which is related to the constituents in leaf) to estimate LNC. Meanwhile, the reflectance factors, derived from photometric ($BRF_{lab-field}$) and polarimetric (IpRF) measurements, have been confirmed to be similar when used to relate to LNC based on the hyperspectral indices. Based on these findings, it would be beneficial to obtain both the photometric and polarimetric properties of light reflected from leaves through only the polarimetric measurements. Thus, the complete description of the optical properties of light reflected from leaves is simplified.

When the BPRF (or specular reflection) dominates the angular reflection of leaves, using the IpRF or $BRF_{lab-field}$ to estimate LNC may lead to a low accuracy in both laboratory and field conditions. NpRF, which is not related to the specular reflection of leaves, can improve the ability of several hyperspectral indices to estimate LNC over a wide range of measurement directions and plant species. LNC estimation from NpRF provides a stable and reliable result for the investigations of plant science. In the future studies, canopy level LNC estimation will be performed based on multiangular, polarimetric, and hyperspectral measurements.

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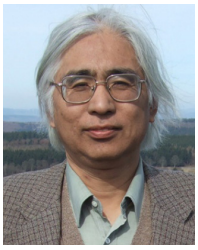
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